

EE 575 : Applied Statistics for Data and Signals

Course Code : EE 575

Course Name : Applied Statistics for Data and Signals

L-T-P-C : 3-0-2-4

Intended for : B.Tech./M.S./M.Tech./Ph.D.

Prerequisite : EE304/ EE503/ EE305/ MA524 or equivalent

Mutual Exclusion: EEXXX (Signals & Systems)

Approval: 54th BoA

Course Contents

- **Mathematical background:** Probability measure space, sample space, σ -algebra, measure theory, random variables, CDF, PDF, joint, marginal and conditional PDFs, transformation of random variable, statistical averages, vector space, inner products, norms, metric, stochastic processes, stationarity, Gaussian process, white noise, stochastic processes through LTI systems. (4 Hours)
- **Statistical Detection Theory:** Binary testing, Decision criteria: Neyman-Pearson Criterion, Bayes Criterion, Min-Max Criterion, receiver operating characteristics (ROC), M-ary decision, erasure decision Bayes risk detectors, sequential detection, Composite hypothesis testing: Bayesian and generalized likelihood ratio test (GLRT), locally most powerful (LMP) detectors, asymptotically equivalent tests. (15 Hours)
- **Estimation for deterministic parameters:** Principle of estimation and its application, properties of estimators. - Minimum Variance Unbiased Estimation: existence and search of MVU estimators, extension to vector parameters. Cramer-Rao Lower Bound: signals in white Gaussian noise, parameter transformation, vector parameters, general Gaussian case. - Linear Models and Unbiased Estimators, scalar and vector Best Linear Unbiased Estimators (BLUE); Maximum Likelihood estimation, Least Squares estimation: linear, order-recursive, sequential, constrained, and nonlinear. (11 Hours)
- **Estimation for random parameters:** Bayesian Estimation: Bayesian linear model, nuisance parameters. Bayesian estimation for deterministic parameters, risk functions, MMSE and MAP estimators, scalar, vector, anti sequential Linear MMSE estimators, Wiener filtering. (6 Hours)
- **Regression analysis:** Simple linear regression, multiple linear regression, estimation and interpretation of regression coefficients, goodness of fit, confidence intervals and hypothesis testing for regression coefficients, generalized linear models. (6 Hours)

Laboratory/practical/tutorial Modules:

- Data preprocessing, Density Estimation methods, feature vector selection, numerical integration methods, ROC

Textbooks:

1. S. M. Kay, **Fundamentals of Statistical Signal Processing, Vol I: Estimation Theory**, Prentice Hall, 1993.



2. H. L. van Trees, K. L. Bell, and Z. Tian, **Detection, Estimation, and Modulation Theory, Part 1: Detection, Estimation, and Filtering Theory**, Wiley, 2013.

References:

1. L. L. Scharf, **Statistical Signal Processing: Detection, Estimation, and Time Series Analysis**, Addison-Wesley, 1991.
2. . H. Y. Poor, **An Introduction to Signal Detection and Estimation**, Springer- Verlag, 1994.
3. C. W. Helstrom, **Elements of Signal Detection and Estimation**, Prentice Hall, 1995.
4. G. Casella and R. L. Berger, **Statistical Inference**, Duxbury Press, 2002.
5. S. M. Kay, **Fundamentals of Statistical Signal Processing, Vol 2: Detection Theory**, Prentice Hall, 1998.